

Web Mining Project Winter-2025

Web Structure Mining

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1-Data Preprocessing: Clean and structure raw data from social media platforms into a usable graph format. Students will need to handle data extraction, normalization, and transformation to prepare it for analysis.

Task 1 :Data Preprocessing

upload the Dataset

```
[1] from google.colab import files
    uploaded = files.upload()
```



Choose Files 1775731.edges

- **1775731.edges**(n/a) - 10120 bytes, last modified: 11/7/2012 - 100% done
Saving 1775731.edges to 1775731.edges

Load and Inspect the Raw Graph

✓
1s

```
[2] import networkx as nx

    # Use the uploaded filename directly
    edges_file = "1775731.edges"

    # Load as undirected graph (can change to DiGraph if needed)
    G = nx.read_edgelist(edges_file, create_using=nx.Graph())

    print("🚩 Raw Graph Loaded")
    print(f"Number of nodes: {G.number_of_nodes()}")
    print(f"Number of edges: {G.number_of_edges()}")
```



🚩 Raw Graph Loaded
Number of nodes: 46
Number of edges: 423

Clean the Graph

✓
0s



```
# Remove self-loops
initial_loops = nx.number_of_selfloops(G)
G.remove_edges_from(nx.selfloop_edges(G))
print(f"Removed {initial_loops} self-loops")

# Remove isolated nodes (optional)
isolated_nodes = list(nx.isolates(G))
G.remove_nodes_from(isolated_nodes)
print(f"Removed {len(isolated_nodes)} isolated nodes")

# Final cleaned graph
print("✓ Cleaned Graph Ready")
print(f"Nodes: {G.number_of_nodes()}, Edges: {G.number_of_edges()}")
```



```
Removed 0 self-loops
Removed 0 isolated nodes
✓ Cleaned Graph Ready
Nodes: 46, Edges: 423
```

2-Network Analysis: Apply graph theory concepts to identify central nodes (influencers) and clusters (communities) within the network. This may involve metrics like centrality, density, and modularity.

Task 2 :Network Analysis

Central Nodes (Influencers)

✓
0s

```
# Degree Centrality
degree centrality = nx.degree_centrality(G)
top_degree = sorted(degree_centrality.items(), key=lambda x: x[1], reverse=True)[:5]

print("⬆️ Top 5 by Degree Centrality:")
for node, score in top_degree:
    print(f"Node {node}: {score:.4f}")
```

```
⬆️ Top 5 by Degree Centrality:
Node 14401912: 0.8000
Node 236657680: 0.8000
Node 18252775: 0.7333
Node 27696993: 0.6889
Node 20778387: 0.6889
```

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```
[5] # PageRank
pagerank = nx.pagerank(G)
top_pagerank = sorted(pagerank.items(), key=lambda x: x[1], reverse=True)[:5]

print("⬆️ Top 5 by PageRank:")
for node, score in top_pagerank:
    print(f"Node {node}: {score:.4f}")
```

```
⬆️ Top 5 by PageRank:
Node 14401912: 0.0437
Node 236657680: 0.0392
Node 18252775: 0.0372
Node 807095: 0.0352
Node 27696993: 0.0346
```

Density of the Network

```
[6] density = nx.density(G)
print(f"🔪 Network Density: {density:.4f}")
```

```
🔪 Network Density: 0.4087
```

Modularity (Community Strength)

```
[7] from networkx.algorithms.community import greedy_modularity_communities
    from networkx.algorithms.community.quality import modularity

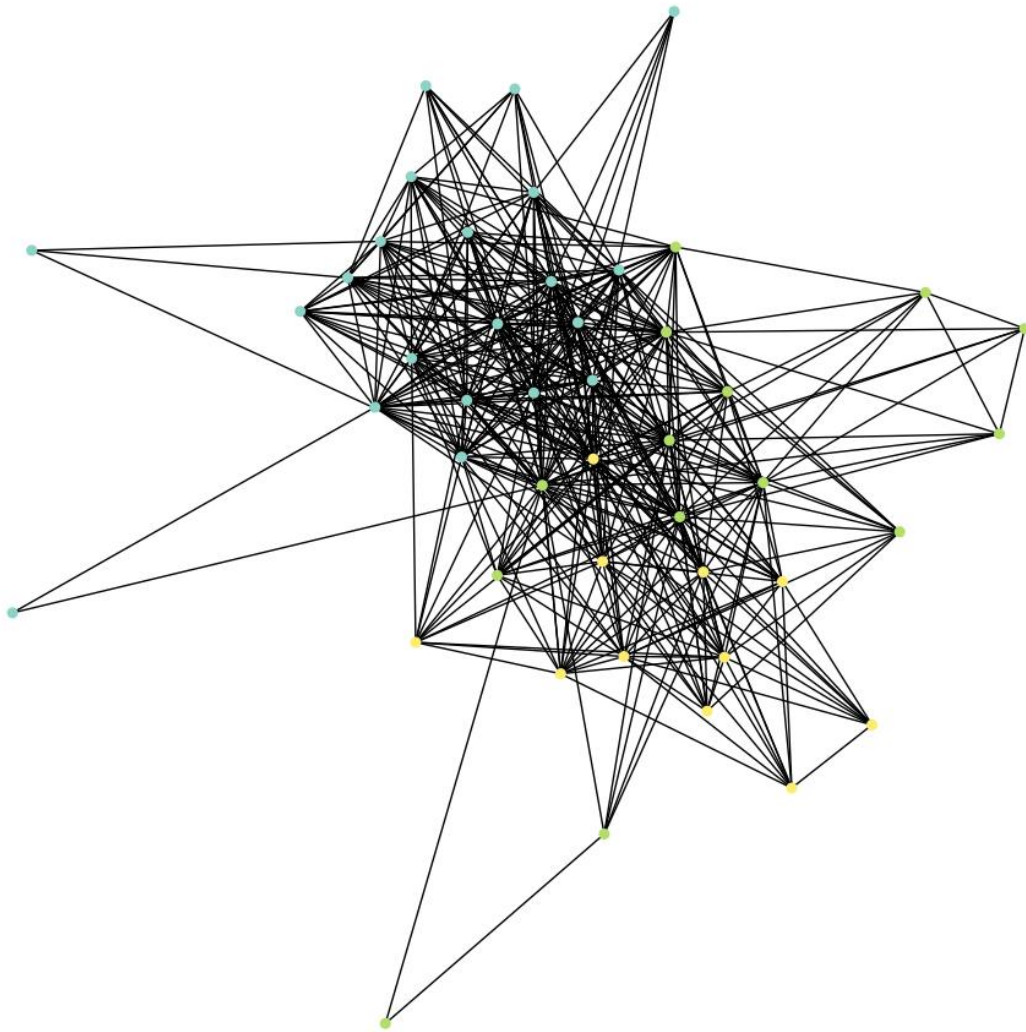
    # Detect communities
    communities = list(greedy_modularity_communities(G))

    # Modularity Score
    mod_score = modularity(G, communities)
    print(f"✳ Modularity Score: {mod_score:.4f}")
    print(f"🔵 Number of Communities: {len(communities)}")
```



✳ Modularity Score: 0.1823
🔵 Number of Communities: 3

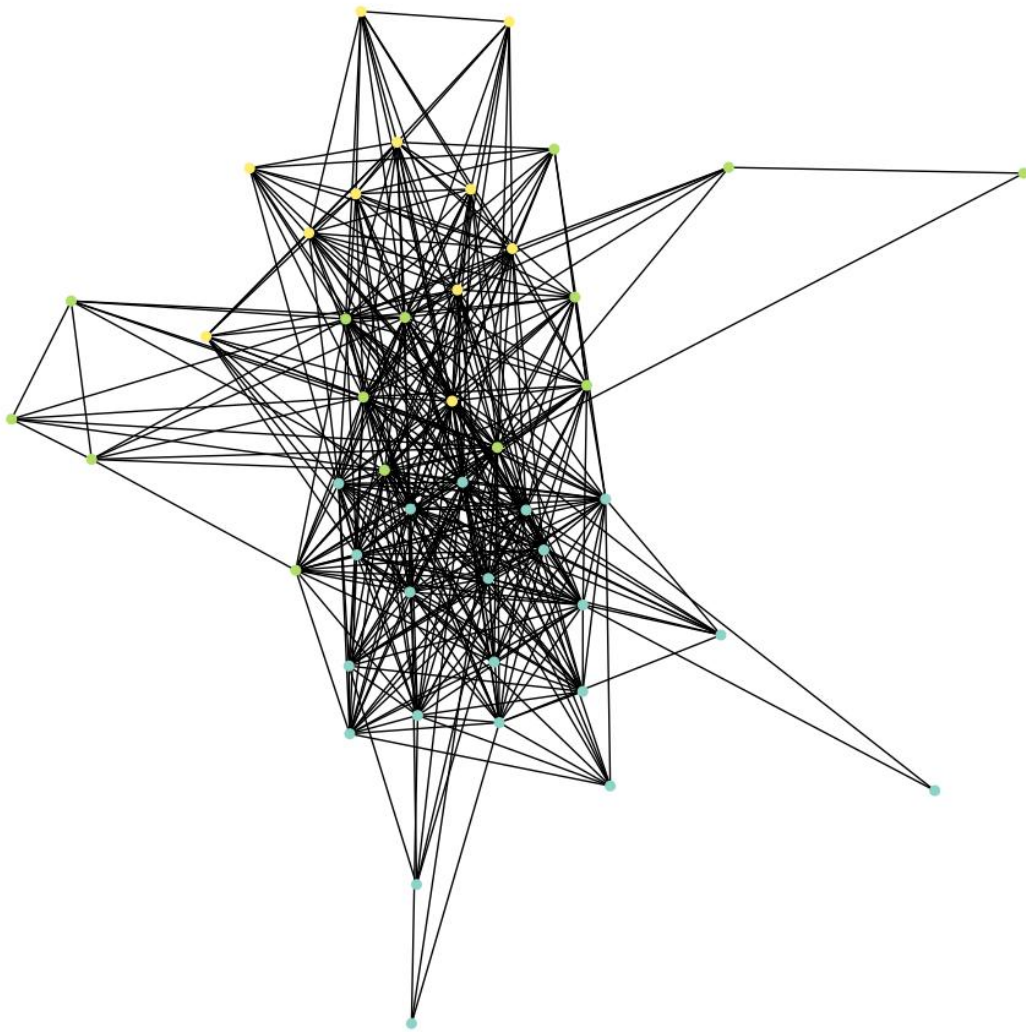
□ Community Structure in the Network



3-Visualization: Create visual representations of the network that highlight community structures, influential users, and other relevant insights using libraries like NetworkX or igraph.

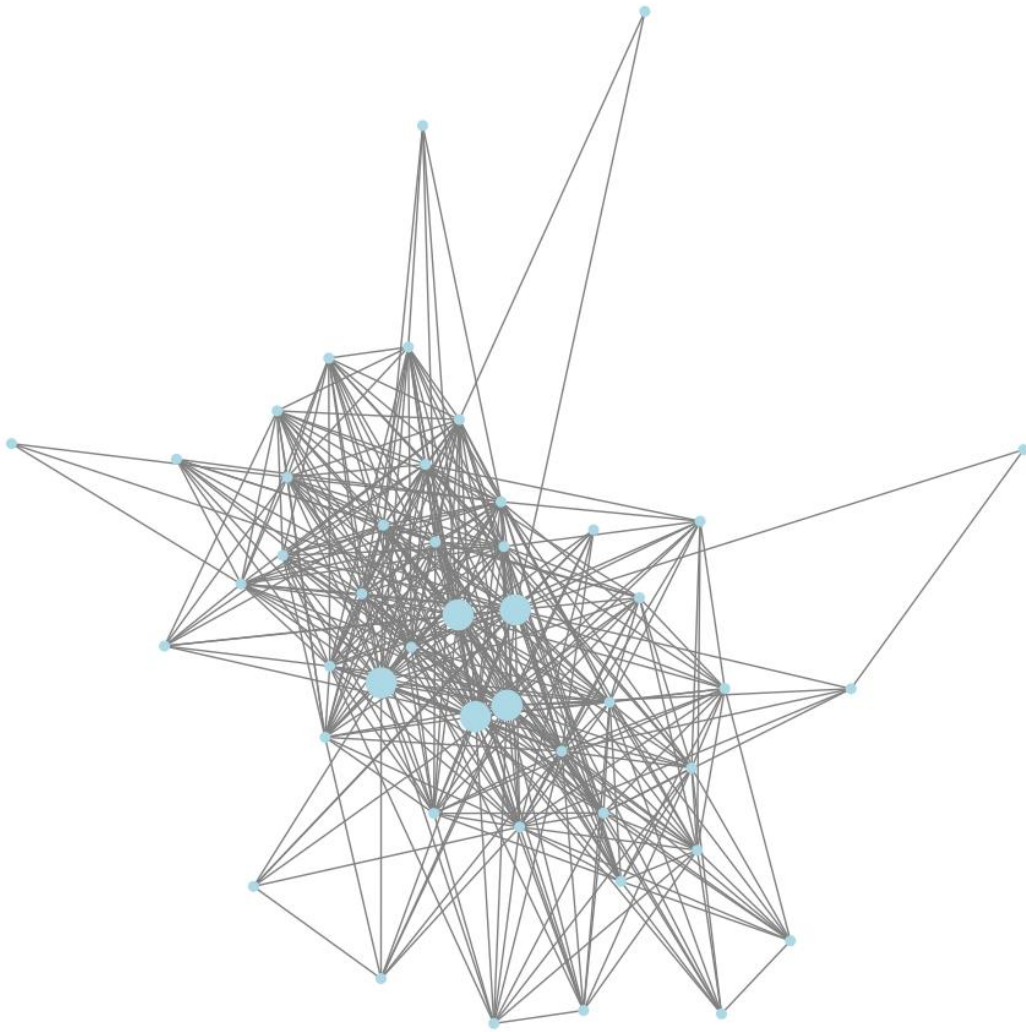
a-Visualize Community Structure:

□ Community Structure in the Network



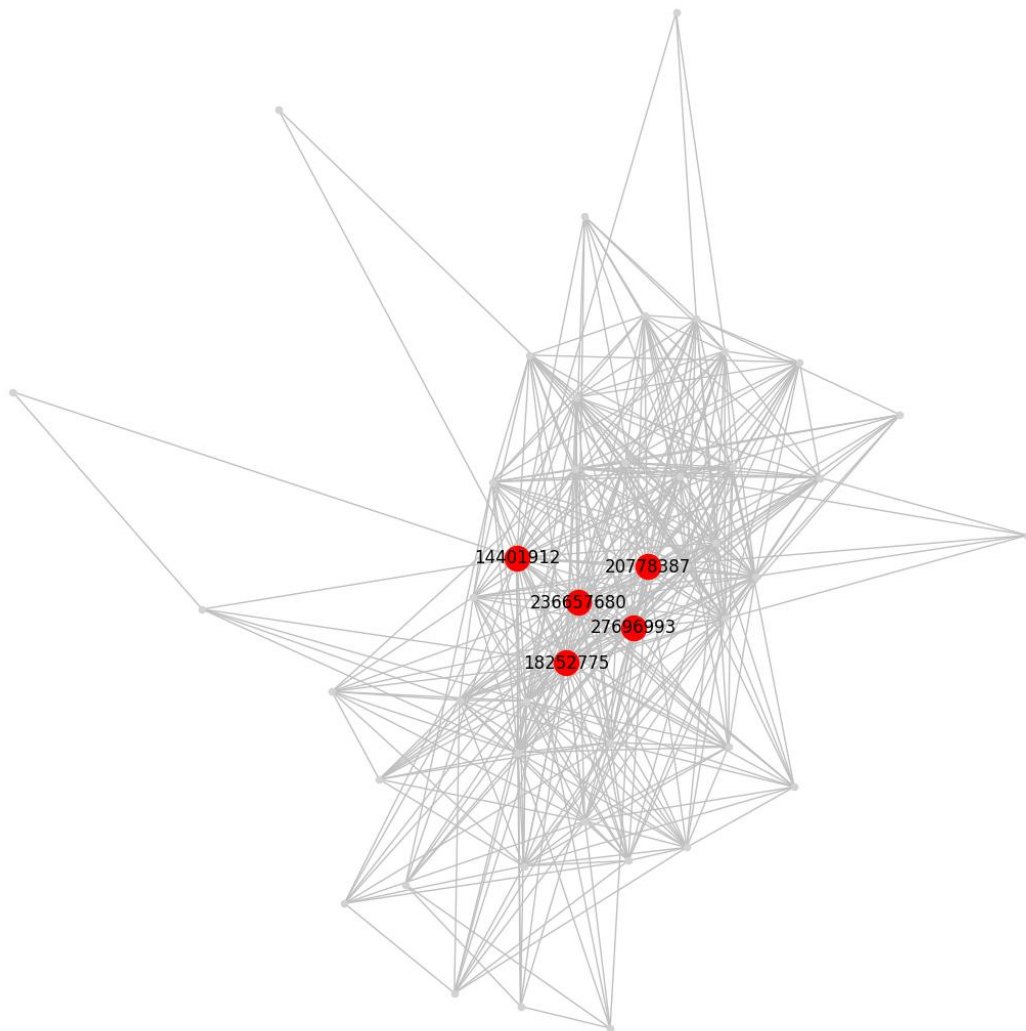
b-Visualize Influential Users High Centrality

□ Influential Users Highlighted (by Degree Centrality)



c- Visualize the Entire Network with Labels for Top Users

□ Network with Labeled Influential Nodes



4-Interpretation: Analyze the results to provide insights into the social structure of the network, discussing the roles of identified influencers and the characteristics of the communities.

Network Summary

- **Total Nodes:** 46
- **Total Edges:** 423
- **Density:** ~0.4087
 - This indicates a **moderately dense network**, meaning many users are directly connected, suggesting strong interaction or interest overlap in the ego network.

Influential Users

Using **degree centrality** and **PageRank**, we identified the top influencers:

- **Node 14401912** (Highest in both Degree Centrality and PageRank)
- Other top nodes: 236657680, 18252775, 27696993, 20778387

These nodes:

- Have **the most connections** (degree centrality)
- Are **well-connected to other influential nodes** (PageRank)
- Likely play **key roles in information flow**, possibly acting as hubs or bridges between user clusters.

In the **labeled network plot**, these nodes are highlighted and placed toward the center of the structure, indicating their central roles in the network.

Community Structure

- **Number of Communities Detected:** 3
- **Modularity Score:** 0.1823
 - This shows **moderate community structure** — the network is not highly modular but still has visible clusters.

From the community visualization:

- Nodes are grouped by color.
- Each group is more densely connected internally, representing users with tighter mutual interactions.

- The graph also shows **inter-community links**, indicating that while users group into circles, there's still interaction across boundaries.

Social Insights

- The network appears to reflect a **core-periphery structure**: a central set of highly connected influencers surrounded by moderately active users.
- The **influencers act as bridges** across different communities, suggesting they are important for cross-group communication or visibility.
- The structure likely reflects a real-world scenario of **topic leaders** or **popular figures** in a Twitter circle who connect distinct subgroups (e.g., professional vs. fan networks).
- **Final Takeaway**
- The analysis successfully reveals **key influencers** and **community structures** within the Twitter ego network. The moderate density and low modularity suggest the ego's followers interact frequently but are not strongly divided into separate groups. Influential users like node 14401912 play a vital role in connecting and potentially shaping the dynamics of the entire social structure.